
Constraint Satisfaction Problem

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Artificial Intelligence

Slides adapted from materials by

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Constraint satisfaction problems (CSPs)

Standard search problem:

state is a “black box”—any old data structure
that supports goal test, eval, successor

CSP:

state is defined by **variables** X_i with **values** from **domain** D_i

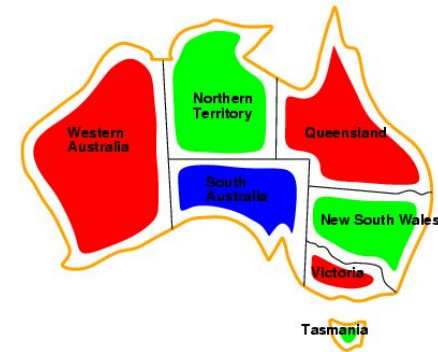
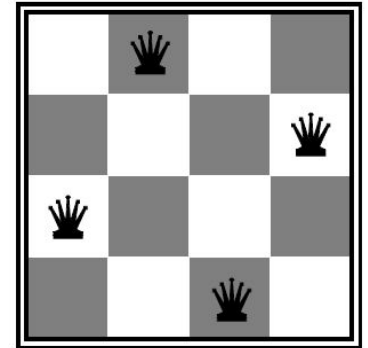
goal test is a set of **constraints** specifying
allowable combinations of values for subsets of variables

Simple example of a **formal representation language**

Allows useful **general-purpose** algorithms with more power
than standard search algorithms

Constraint Satisfaction Problems

- Standard search problems:
 - State is a “black box”: arbitrary data structure
 - Goal test: any function over states
 - Successor function can be anything
- Constraint satisfaction problems (CSPs):
 - A special subset of search problems
 - State is defined by variables X_i with values from a domain D (sometimes D depends on i)
 - Goal test is a set of constraints specifying allowable combinations of values for subsets of variables
 - We call the set of states that are induced by a set of variables the set of possible worlds



Constraint Satisfaction Problems

- Simple example of a formal representation language
- Allows useful general-purpose algorithms with more power than standard search algorithms

Examples

- Crossword Puzzle:
 - variables are words that have to be filled in
 - domains are English words of the correct length
 - Possible worlds: all ways of assigning words
- Crossword 2:
 - variables are cells (individual squares)
 - domains are letters of the alphabet
 - possible worlds: all ways of assigning letters to cells
- Sudoku:
 - variables are cells
 - domains are numbers between 1 and 9
 - possible worlds: all ways of assigning numbers to cells

Examples

- Scheduling Problem:
 - variables are different tasks that need to be scheduled (e.g., course in a university; job in a machine shop)
 - domains are the different combinations of times and locations
 - for each task (e.g., time/room for course; time/machine for job)
 - possible worlds: time/location assignments for each task
- n-Queens problem":
 - variable: location of a queen on a chess board
 - there are n of them in total, hence the name
 - domains: grid coordinates
 - possible worlds: locations of all queens

Constraints

Constraints are restrictions on the values that one or more variables can take

- Unary constraint: restriction involving a single variable
- k-ary constraint: restriction involving the domains of k different variables
 - k-ary constraints can be represented as binary constraints
- Constraints can be specified by
 - giving a list of valid domain values for each variable participating in the constraint
 - giving a function that returns true when given values for each variable which satisfy the constraint
- A possible world satisfies a set of constraints if the set of variables involved in each constraint take values that are consistent with that constraint

CSP Problem Definition

- A constraint satisfaction problem consists of:
 - a set of variables
 - a domain for each variable
 - a set of constraints
- A model of a CSP is an assignment of values to variables that satisfies all of the constraints.

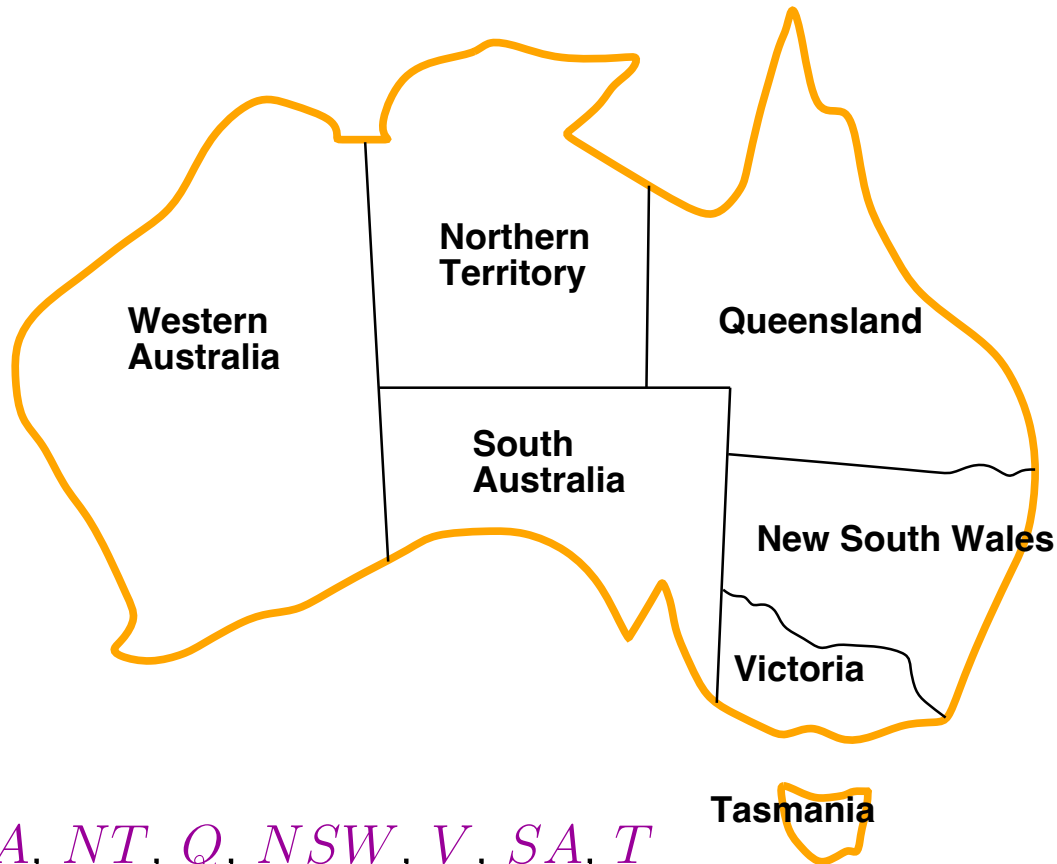
Examples

- Crossword Puzzle:
 - variables are words that have to be filled in domains are valid English words
 - constraints: words have the same letters at points where they intersect
- Crossword 2:
 - variables are cells (individual squares)
 - domains are letters of the alphabet
 - constraints: sequences of letters form valid English words
- Sudoku
 - variables are cells
 - domains are numbers between 1 and 9
 - constraints: rows, columns, boxes contain all different numbers

Examples

- Scheduling Problem:
 - variables are different tasks that need to be scheduled (e.g., course in a university; job in a machine shop)
 - domains are the different combinations of times and locations for each task (e.g., time/room for course; time/machine for job)
 - constraints: tasks can't be scheduled in the same location at the same time; certain tasks can't be scheduled in different locations at the same time; some tasks must come earlier than others; etc.
- n-Queens problem:
 - variable: location of a queen on a chess board
 - domains: grid coordinates
 - constraints: no queen can attack another

Example: Map-Coloring



Variables WA, NT, Q, NSW, V, SA, T

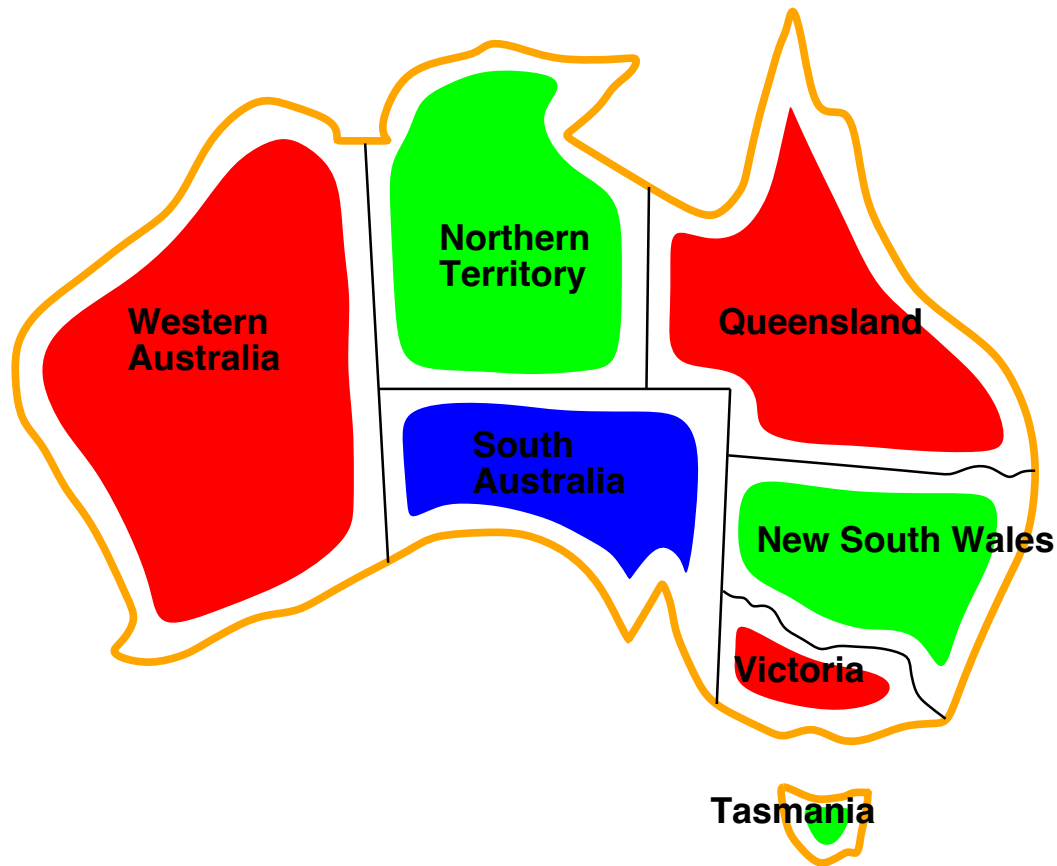
Domains $D_i = \{red, green, blue\}$

Constraints: adjacent regions must have different colors

e.g., $WA \neq NT$ (if the language allows this), or

$(WA, NT) \in \{(red, green), (red, blue), (green, red), (green, blue), \dots\}$

Example: Map-Coloring contd.



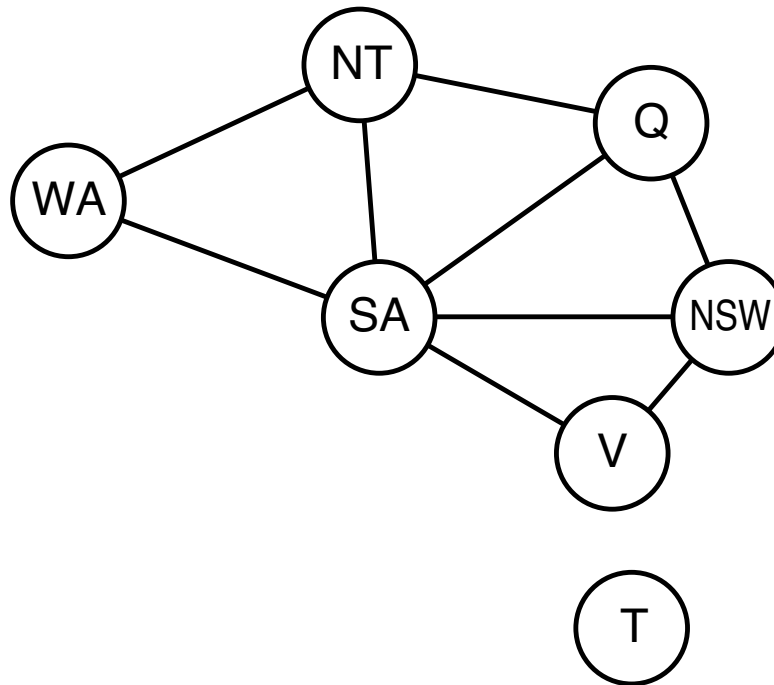
Solutions are assignments satisfying all constraints, e.g.,

$\{WA = red, NT = green, Q = red, NSW = green, V = red, SA = blue, T = green\}$

Constraint graph

Binary CSP: each constraint relates at most two variables

Constraint graph: nodes are variables, arcs show constraints



General-purpose CSP algorithms use the graph structure to speed up search. E.g., Tasmania is an independent subproblem!

Varieties of CSPs

Discrete variables

finite domains; size $d \Rightarrow O(d^n)$ complete assignments

- ◇ e.g., Boolean CSPs, incl. Boolean satisfiability (NP-complete)

infinite domains (integers, strings, etc.)

- ◇ e.g., job scheduling, variables are start/end days for each job
- ◇ need a **constraint language**, e.g., $StartJob_1 + 5 \leq StartJob_3$
- ◇ **linear** constraints solvable, **nonlinear** undecidable

Continuous variables

- ◇ e.g., start/end times for Hubble Telescope observations
- ◇ linear constraints solvable in poly time by LP methods

Varieties of constraints

Unary constraints involve a single variable,

e.g., $SA \neq \text{green}$

Binary constraints involve pairs of variables,

e.g., $SA \neq WA$

Higher-order constraints involve 3 or more variables,

e.g., cryptarithmic column constraints

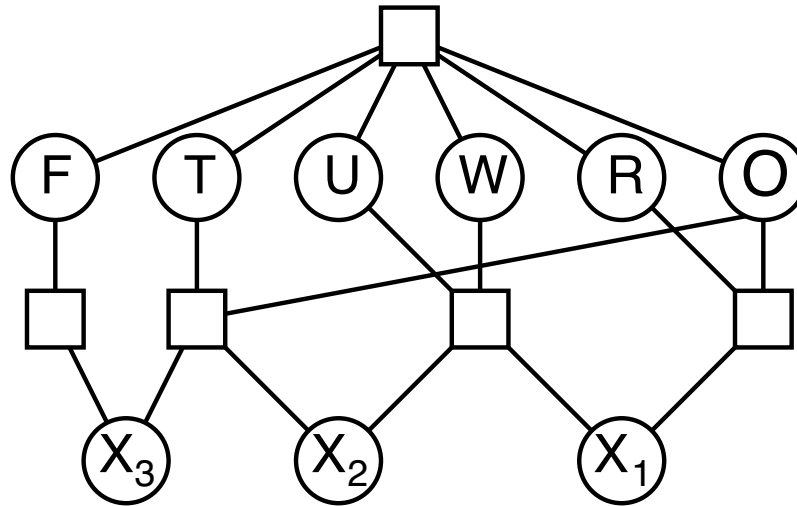
Preferences (soft constraints), e.g., red is better than green

often representable by a cost for each variable assignment

→ constrained optimization problems

Example: Cryptarithmic

$$\begin{array}{r}
 T \ W \ O \\
 + \ T \ W \ O \\
 \hline
 F \ O \ U \ R
 \end{array}$$



Variables: $F \ T \ U \ W \ R \ O \ X_1 \ X_2 \ X_3$

Domains: $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

Constraints

$alldiff(F, T, U, W, R, O)$

$O + O = R + 10 \cdot X_1$, etc.

Real-world CSPs

Assignment problems

e.g., who teaches what class

Timetabling problems

e.g., which class is offered when and where?

Hardware configuration

Spreadsheets

Transportation scheduling

Factory scheduling

Floorplanning

Notice that many real-world problems involve real-valued variables

Standard search formulation (incremental)

Let's start with the straightforward, dumb approach, then fix it

States are defined by the values assigned so far

- ◇ **Initial state:** the empty assignment, $\{ \}$
- ◇ **Successor function:** assign a value to an unassigned variable that does not conflict with current assignment.
 $\Rightarrow$ fail if no legal assignments (not fixable!)
- ◇ **Goal test:** the current assignment is complete

- 1) This is the same for all CSPs! 😊
- 2) Every solution appears at depth n with n variables
 $\Rightarrow$ use depth-first search
- 3) Path is irrelevant, so can also use complete-state formulation
- 4) $b = (n - \ell)d$ at depth ℓ , hence $n!d^n$ leaves!!!! 😞

Backtracking search

Variable assignments are *commutative*, i.e.,

$[WA = \text{red} \text{ then } NT = \text{green}]$ same as $[NT = \text{green} \text{ then } WA = \text{red}]$

Only need to consider assignments to a single variable at each node

$\Rightarrow b = d$ and there are d^n leaves

Depth-first search for CSPs with single-variable assignments is called *backtracking* search

Backtracking search is the basic uninformed algorithm for CSPs

Can solve n -queens for $n \approx 25$

Backtracking search

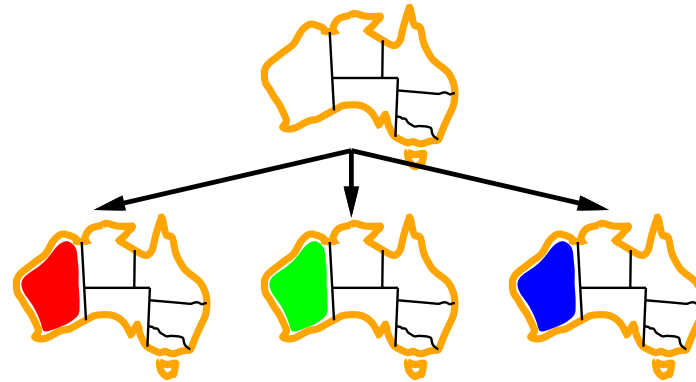
```
function BACKTRACKING-SEARCH(csp) returns solution/failure
  return RECURSIVE-BACKTRACKING({ }, csp)

function RECURSIVE-BACKTRACKING(assignment, csp) returns soln/failure
  if assignment is complete then return assignment
  var ← SELECT-UNASSIGNED-VARIABLE(VARIABLES[csp], assignment, csp)
  for each value in ORDER-DOMAIN-VALUES(var, assignment, csp) do
    if value is consistent with assignment given CONSTRAINTS[csp] then
      add {var = value} to assignment
      result ← RECURSIVE-BACKTRACKING(assignment, csp)
      if result ≠ failure then return result
      remove {var = value} from assignment
  return failure
```

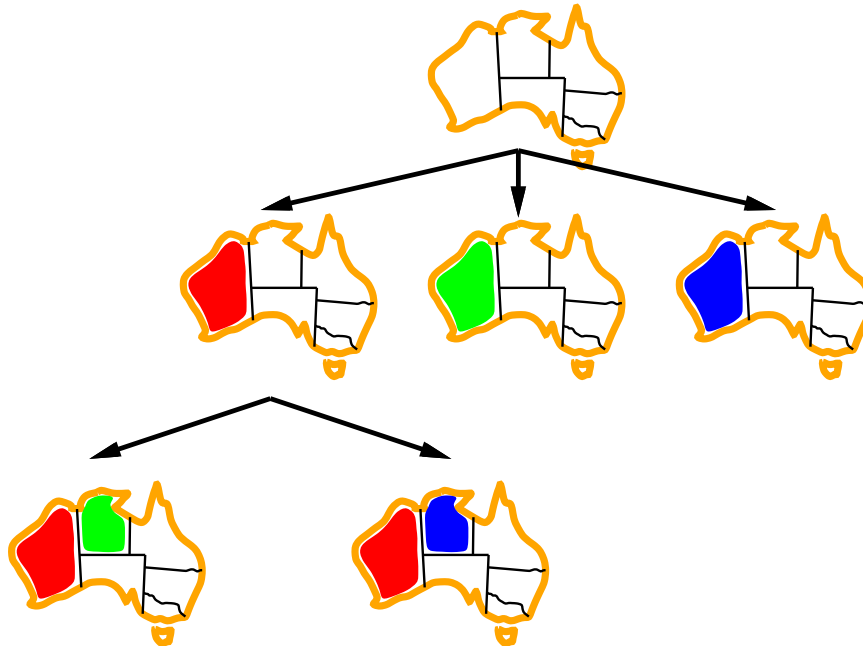
Backtracking example



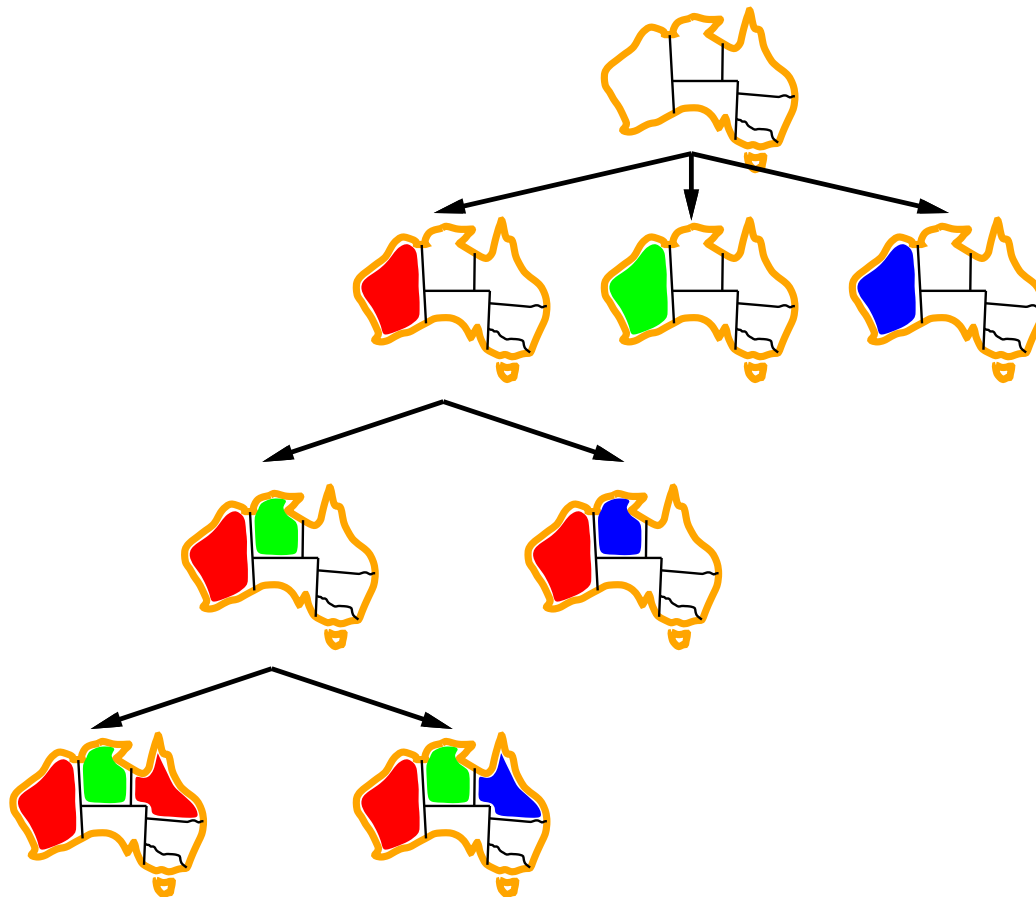
Backtracking example



Backtracking example



Backtracking example



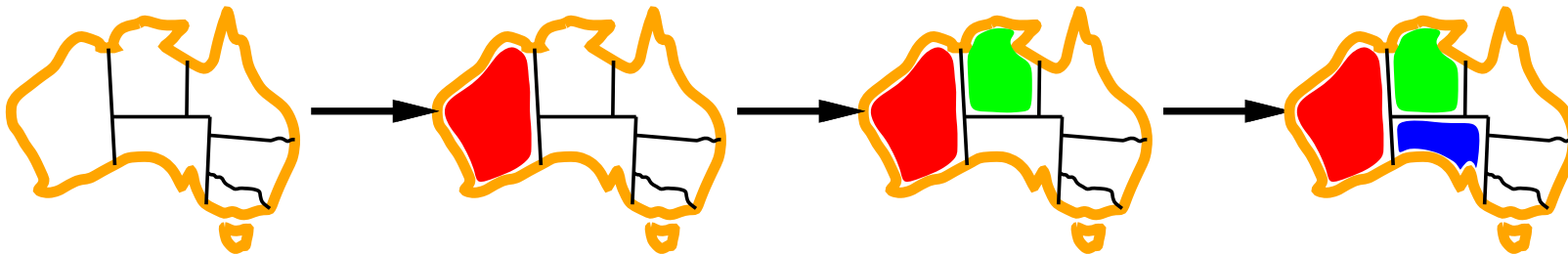
Improving backtracking efficiency

General-purpose methods can give huge gains in speed:

1. Which variable should be assigned next?
2. In what order should its values be tried?
3. Can we detect inevitable failure early?
4. Can we take advantage of problem structure?

Minimum remaining values

Minimum remaining values (MRV):
choose the variable with the fewest legal values



Which variable should be assigned next?

□ minimum remaining values heuristic

- Most constrained variable:



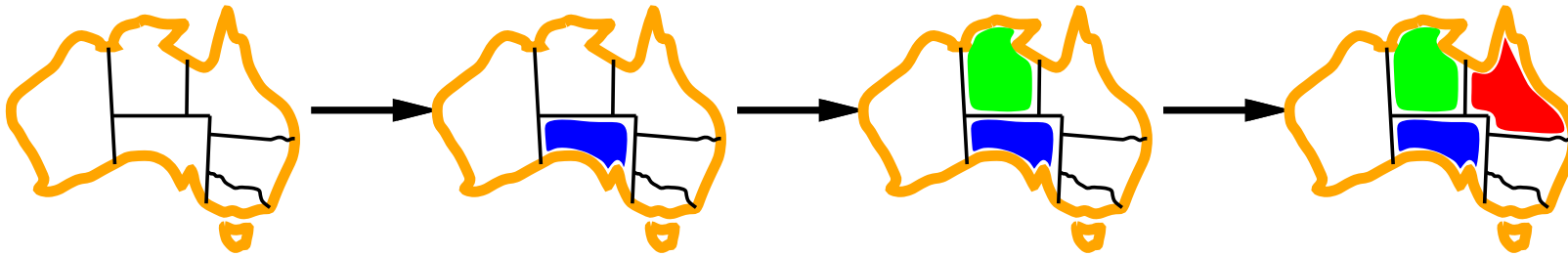
- a.k.a. *minimum remaining values (MRV)* heuristic
- Picks a variable which will cause failure as soon as possible, allowing the tree to be pruned.

Degree heuristic

Tie-breaker among MRV variables

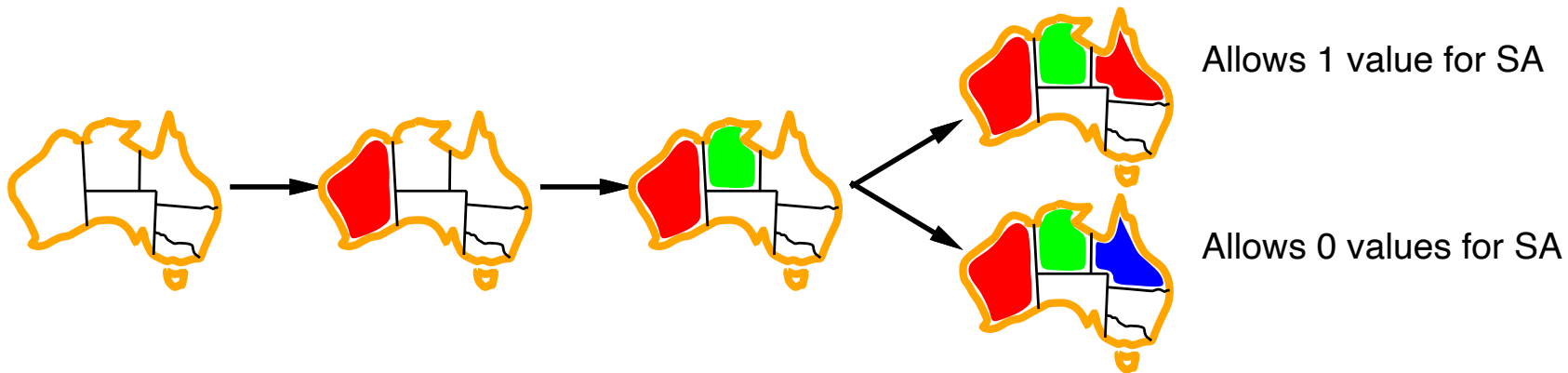
Degree heuristic:

choose the variable with the most constraints on remaining variables



Least constraining value

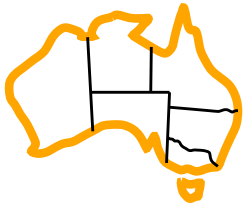
Given a variable, choose the least constraining value:
the one that rules out the fewest values in the remaining variables



Combining these heuristics makes 1000 queens feasible

Forward checking

Idea: Keep track of remaining legal values for unassigned variables
Terminate search when any variable has no legal values



WA

NT

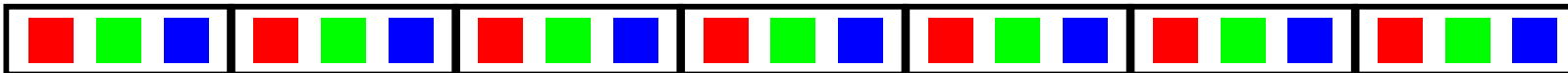
Q

NSW

V

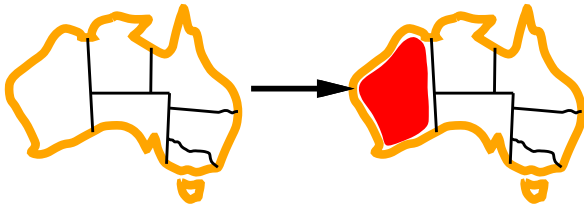
SA

T



Forward checking

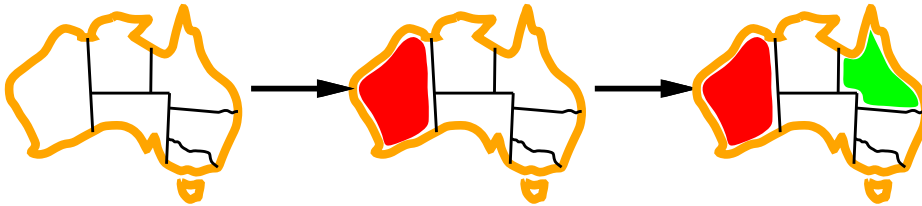
Idea: Keep track of remaining legal values for unassigned variables
Terminate search when any variable has no legal values



WA	NT	Q	NSW	V	SA	T
■ ■ ■	■ ■ ■	■ ■ ■	■ ■ ■	■ ■ ■	■ ■ ■	■ ■ ■
■	■ ■	■ ■ ■	■ ■ ■	■ ■ ■	■ ■	■ ■ ■

Forward checking

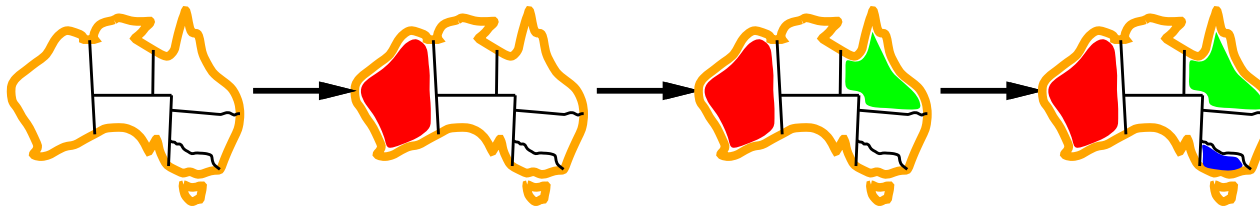
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WA	NT	Q	NSW	V	SA	T
■ ■ ■	■ ■ ■	■ ■ ■	■ ■ ■	■ ■ ■	■ ■ ■	■ ■ ■
■	■ ■	■ ■ ■	■ ■ ■	■ ■ ■	■ ■	■ ■ ■
■	■	■ ■ ■	■ ■	■ ■ ■	■	■ ■ ■

Forward checking

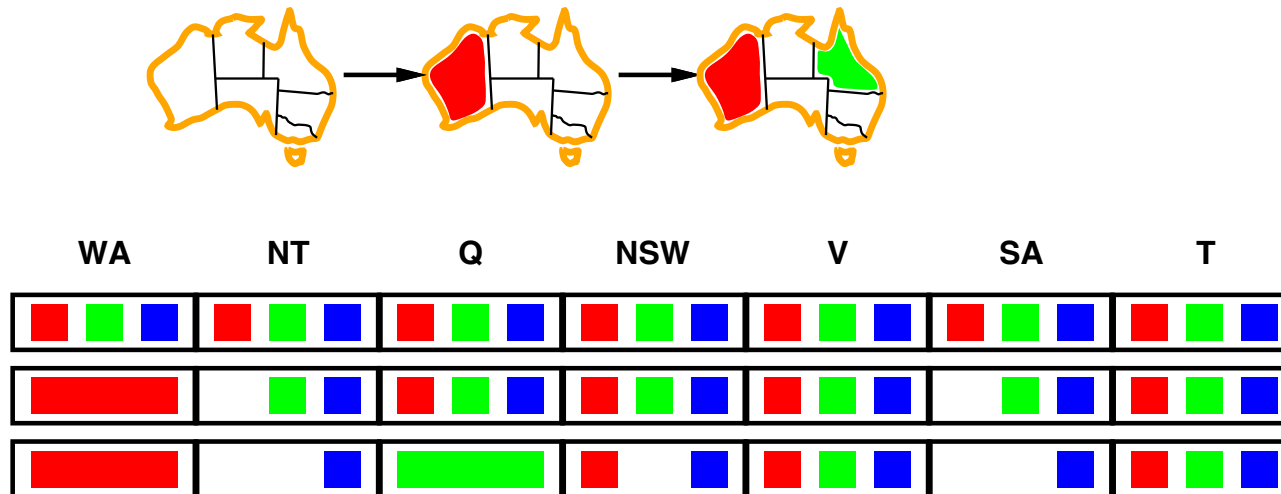
Idea: Keep track of remaining legal values for unassigned variables
 Terminate search when any variable has no legal values



WA	NT	Q	NSW	V	SA	T

Constraint propagation

Forward checking propagates information from assigned to unassigned variables, but doesn't provide early detection for all failures:



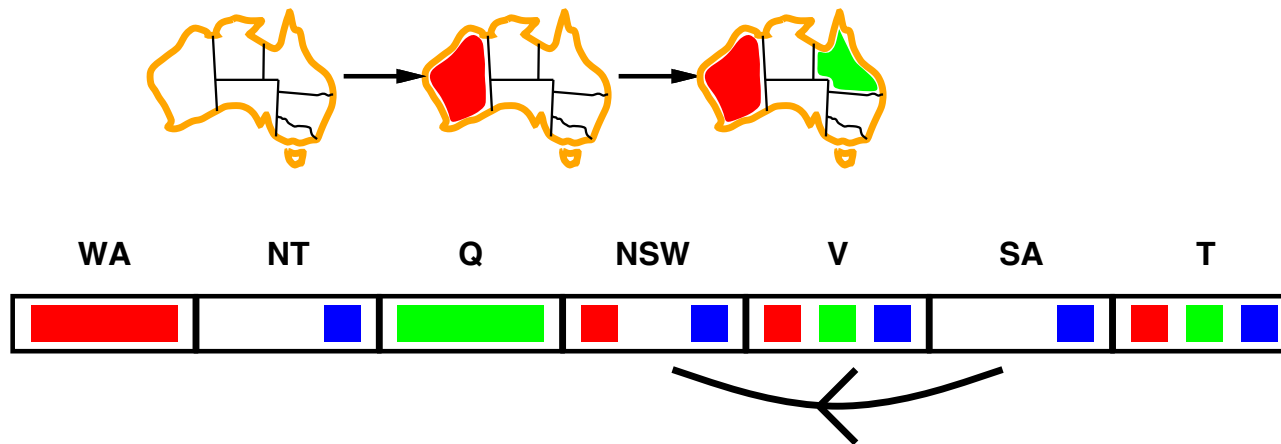
NT and *SA* cannot both be blue!

Constraint propagation repeatedly enforces constraints locally

Arc consistency

Simplest form of propagation makes each arc **consistent**

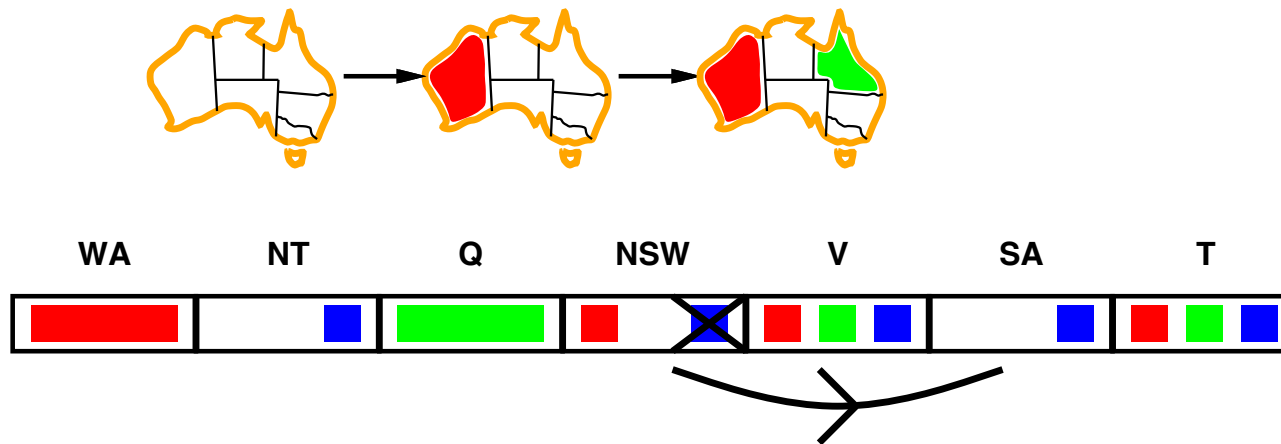
$X \rightarrow Y$ is consistent iff
for **every** value x of X there is **some** allowed y



Arc consistency

Simplest form of propagation makes each arc **consistent**

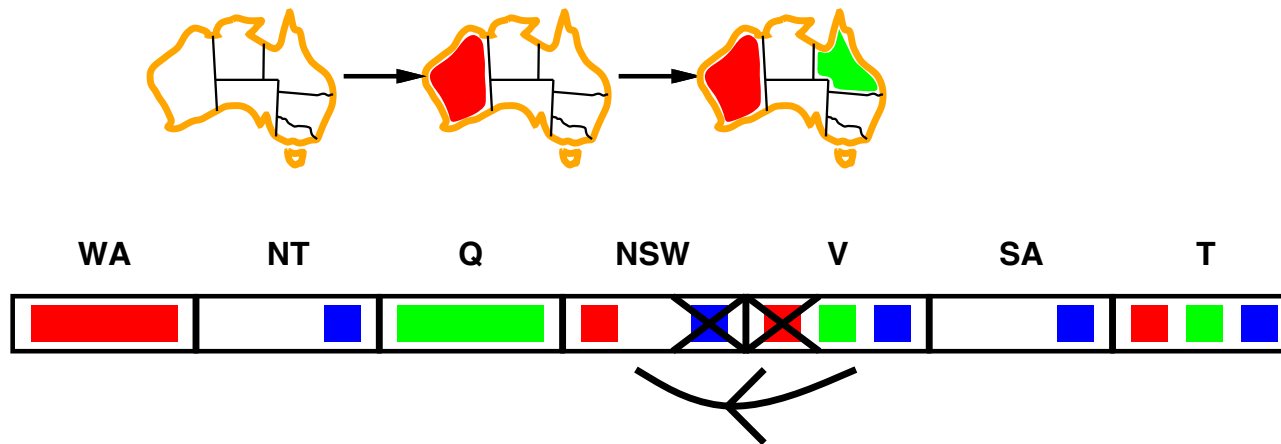
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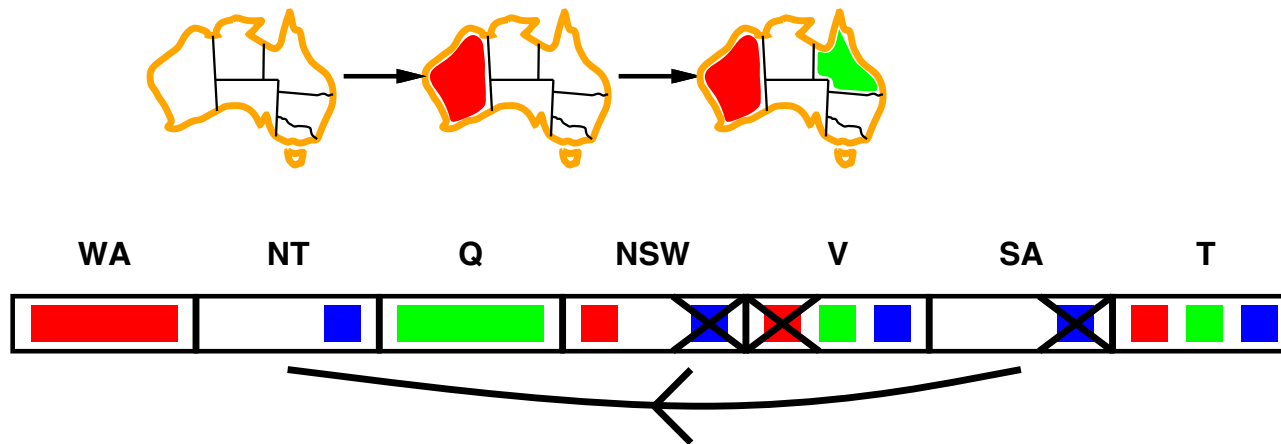


If X loses a value, neighbors of X need to be rechecked

Arc consistency

Simplest form of propagation makes each arc **consistent**

$X \rightarrow Y$ is consistent iff
for **every** value x of X there is **some** allowed y



If X loses a value, neighbors of X need to be rechecked

Arc consistency detects failure earlier than forward checking

Can be run as a preprocessor or after each assignment

Arc consistency algorithm

function AC-3(*csp*) **returns** the CSP, possibly with reduced domains

inputs: *csp*, a binary CSP with variables $\{X_1, X_2, \dots, X_n\}$

local variables: *queue*, a queue of arcs, initially all the arcs in *csp*

while *queue* is not empty **do**

$(X_i, X_j) \leftarrow \text{REMOVE-FIRST}(\textit{queue})$

if REMOVE-INCONSISTENT-VALUES(X_i, X_j) **then**

for each X_k **in** NEIGHBORS[X_i] **do**

 add (X_k, X_i) to *queue*

function REMOVE-INCONSISTENT-VALUES(X_i, X_j) **returns** true iff succeeds

removed $\leftarrow$ false

for each x **in** DOMAIN[X_i] **do**

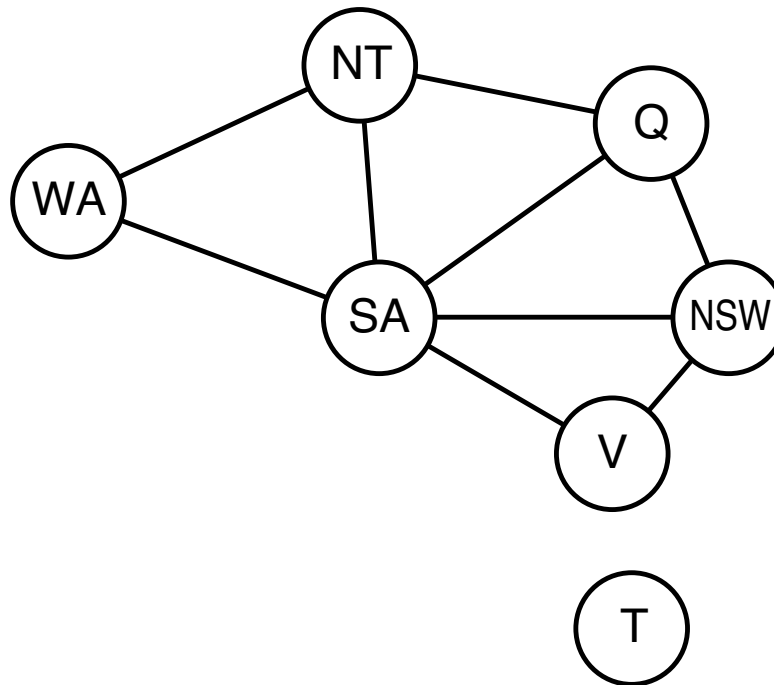
if no value y in DOMAIN[X_j] allows (x, y) to satisfy the constraint $X_i \leftrightarrow X_j$

then delete x from DOMAIN[X_i]; *removed* $\leftarrow$ true

return *removed*

$O(n^2 d^3)$, can be reduced to $O(n^2 d^2)$ (but detecting **all** is NP-hard)

Problem structure



Tasmania and mainland are **independent subproblems**

Identifiable as **connected components** of constraint graph

Problem structure contd.

Suppose each subproblem has c variables out of n total

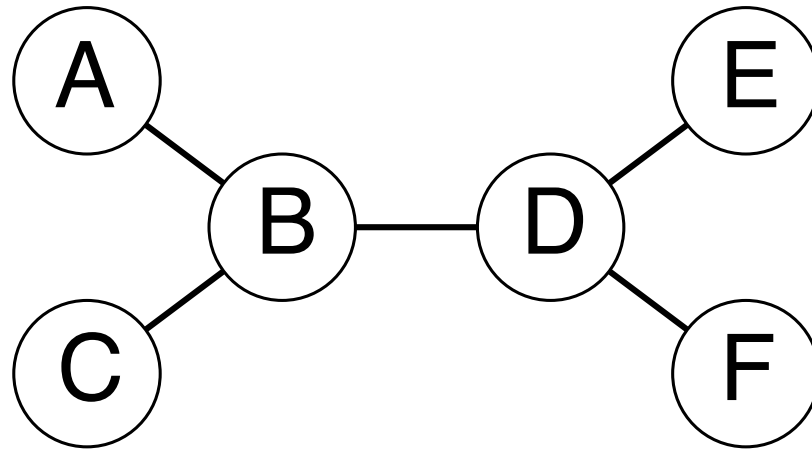
Worst-case solution cost is $n/c \cdot d^c$, **linear** in n

E.g., $n = 80$, $d = 2$, $c = 20$

$2^{80} = 4$ billion years at 10 million nodes/sec

$4 \cdot 2^{20} = 0.4$ seconds at 10 million nodes/sec

Tree-structured CSPs



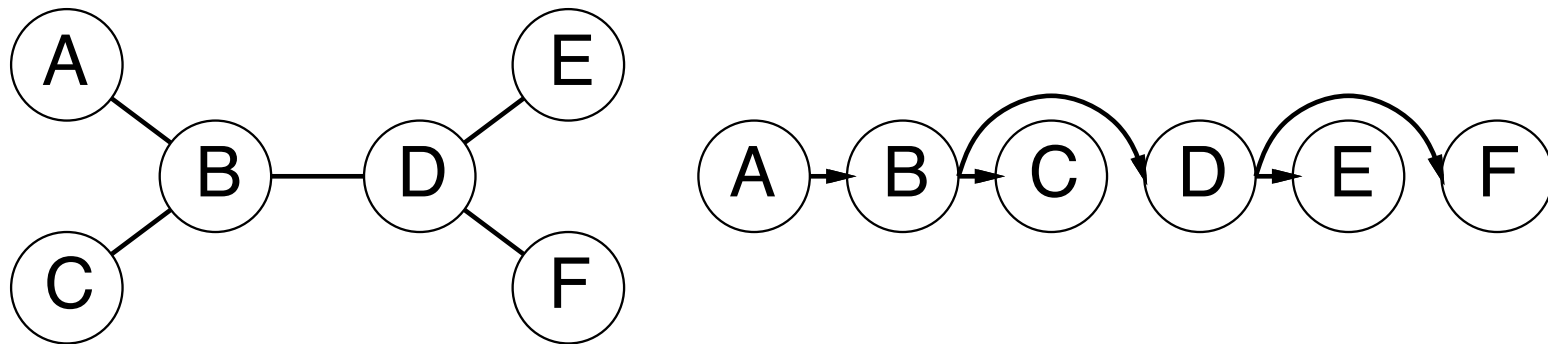
Theorem: if the constraint graph has no loops, the CSP can be solved in $O(n d^2)$ time

Compare to general CSPs, where worst-case time is $O(d^n)$

This property also applies to logical and probabilistic reasoning:
an important example of the relation between syntactic restrictions
and the complexity of reasoning.

Algorithm for tree-structured CSPs

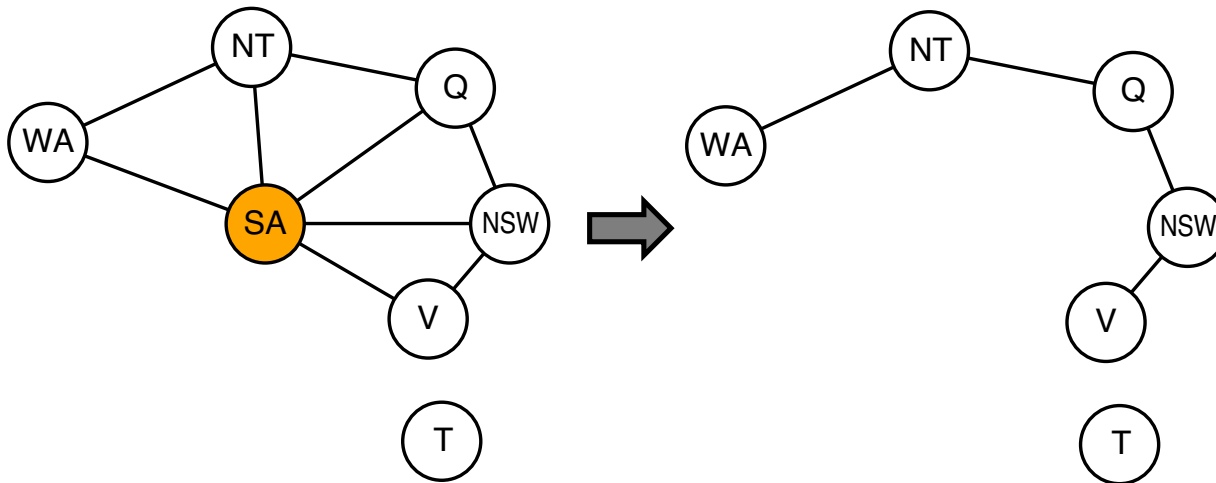
1. Choose a variable as root, order variables from root to leaves such that every node's parent precedes it in the ordering



2. For j from n down to 2, apply $\text{REMOVEINCONSISTENT}(\text{Parent}(X_j), X_j)$
3. For j from 1 to n , assign X_j consistently with $\text{Parent}(X_j)$

Nearly tree-structured CSPs

Conditioning: instantiate a variable, prune its neighbors' domains



Cutset conditioning: instantiate (in all ways) a set of variables such that the remaining constraint graph is a tree

Cutset size $c \Rightarrow$ runtime $O(d^c \cdot (n - c)d^2)$, very fast for small c

Iterative algorithms for CSPs

Hill-climbing, simulated annealing typically work with “complete” states, i.e., all variables assigned

To apply to CSPs:

- allow states with unsatisfied constraints
- operators **reassign** variable values

Variable selection: randomly select any conflicted variable

Value selection by **min-conflicts** heuristic:

- choose value that violates the fewest constraints
- i.e., hillclimb with $h(n)$ = total number of violated constraints

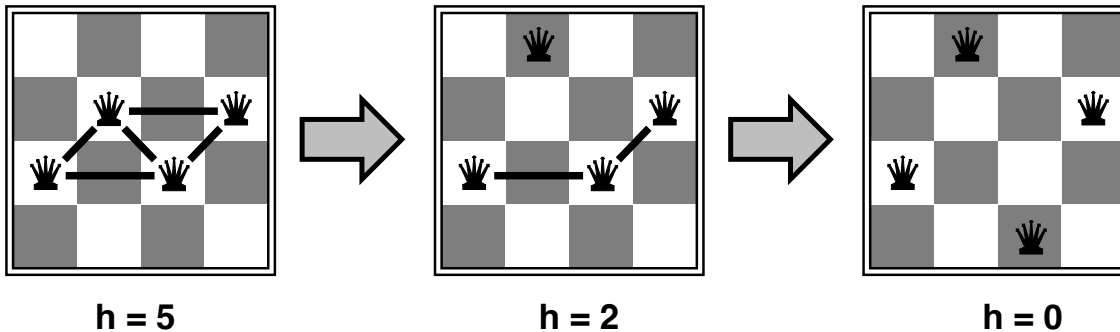
Example: 4-Queens

States: 4 queens in 4 columns ($4^4 = 256$ states)

Operators: move queen in column

Goal test: no attacks

Evaluation: $h(n)$ = number of attacks

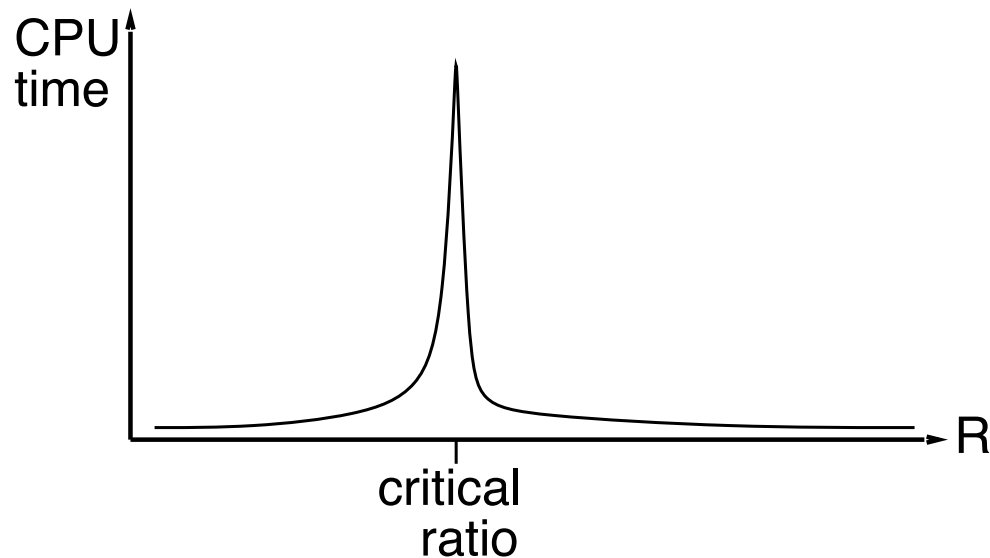


Performance of min-conflicts

Given random initial state, can solve n -queens in almost constant time for arbitrary n with high probability (e.g., $n = 10,000,000$)

The same appears to be true for any randomly-generated CSP **except** in a narrow range of the ratio

$$R = \frac{\text{number of constraints}}{\text{number of variables}}$$



Summary

CSPs are a special kind of problem:

- states defined by values of a fixed set of variables

- goal test defined by **constraints** on variable values

Backtracking = depth-first search with one variable assigned per node

Variable ordering and value selection heuristics help significantly

Forward checking prevents assignments that guarantee later failure

Constraint propagation (e.g., arc consistency) does additional work to constrain values and detect inconsistencies

The CSP representation allows analysis of problem structure

Tree-structured CSPs can be solved in linear time

Iterative min-conflicts is usually effective in practice